

Article

Effects of Land Use Changes on CO₂ Emission Dynamics in the Amazon

Adriano Maltezo da Rocha ¹, Mauricio Franceschi ² , Alan Rodrigo Panosso ¹ ,
Marco Antonio Camillo de Carvalho ³ , Mara Regina Moitinho ¹, Marcílio Vieira Martins Filho ¹ ,
Dener Marcio da Silva Oliveira ^{4,5}, Diego Antonio França de Freitas ⁴ , Oscar Mitsuo Yamashita ³ 
and Newton La Scala, Jr. ^{1,*}

- ¹ School of Agricultural and Veterinary Sciences, São Paulo State University (FCAV–UNESP), Via de Acesso Prof. Paulo Donato Castellane, s/n, Jaboticabal 14884-900, SP, Brazil; admr.maltezo@hotmail.com (A.M.d.R.); alan.panosso@unesp.br (A.R.P.); maramoitinho@gmail.com (M.R.M.); marcilio.martins-filho@unesp.br (M.V.M.F.)
 - ² Postgraduate Program in Tropical Agriculture, Federal University of Mato Grosso, Avenida Fernando Corrêa da Costa, 2367, Cuiabá 78060-900, MT, Brazil; mauriciofranceschi123@hotmail.com
 - ³ Department of Agronomy, Mato Grosso State University (UNEMAT), Rodovia MT 208, km 147, Alta Floresta 78580-000, MT, Brazil; marcocarvalho@unemat.br (M.A.C.d.C.); yama@unemat.br (O.M.Y.)
 - ⁴ Agricultural Sciences Institute, Campus Florestal, Federal University of Viçosa (UFV), Rodovia LMG 818, km 06, Florestal 35690-000, MG, Brazil; dener.oliveira@ufv.br (D.M.d.S.O.); diegofranca@ufv.br (D.A.F.d.F.)
 - ⁵ Research Centre for Greenhouse Gas Innovation (RCGI), Av. Professor Mello Moraes, 2231, Cidade Universitária, São Paulo 05508-030, SP, Brazil
- * Correspondence: la.scale@unesp.br

Abstract: Global climate change is closely tied to CO₂ emissions, and implementing conservation-agricultural systems can help mitigate emissions in the Amazon. By maintaining forest cover and integrating sustainable agricultural practices in pasture, these systems help mitigate climate change and preserve the carbon stocks in Amazon forest soils. In addition, these systems improve soil health, microclimate regulation, and promote sustainable agricultural practices in the Amazon region. This study aimed to evaluate the CO₂ emission dynamics and its relationship with soil attributes under different uses in the Amazon. The experiment consisted of four treatments (Degraded Pasture—DP; Managed Pasture—MP; Native Forest—NF; and Livestock Forest Integration—LF), with 25 replications. Soil CO₂ emission (FCO₂), soil temperature, and soil moisture were evaluated over a period of 114 days, and the chemical, physical, and biological attributes of the soil were measured at the end of this period. The mean FCO₂ reached values of 4.44, 3.88, 3.80, and 3.14 μmol m^{−2} s^{−1} in DP, MP, NF, and LF, respectively. In addition to the direct relationship between soil CO₂ emissions and soil temperature for all land uses, soil bulk density indirectly influenced emissions in NF. The amount of humic acid induced the highest emission in DP. Soil organic carbon and carbon stock were higher in MP and LF. These values demonstrate that FCO₂ was influenced by the Amazon land uses and highlight LF as a low CO₂ emission system with a higher potential for carbon stock in the soil.

Keywords: carbon stock; greenhouse gases; degraded pasture; livestock forest; soil respiration



Academic Editor: Shicheng Li

Received: 30 December 2024

Revised: 28 January 2025

Accepted: 14 February 2025

Published: 18 February 2025

Citation: Rocha, A.M.d.; Franceschi, M.; Panosso, A.R.; Carvalho, M.A.C.d.; Moitinho, M.R.; Martins Filho, M.V.; Oliveira, D.M.d.S.; Freitas, D.A.F.d.; Yamashita, O.M.; La Scala, N., Jr. Effects of Land Use Changes on CO₂ Emission Dynamics in the Amazon. *Agronomy* **2025**, *15*, 488. <https://doi.org/10.3390/agronomy15020488>

Copyright: © 2025 by the authors.

Licensee MDPI, Basel, Switzerland.

This article is an open access article distributed under the terms and conditions of the Creative Commons Attribution (CC BY) license (<https://creativecommons.org/licenses/by/4.0/>).

1. Introduction

Brazil has a prominent position in the world ranking regarding historical emissions of greenhouse gases (CO₂, N₂O, and CH₄), standing out in fourth place among the largest emitters [1]. Research into emissions of greenhouse gases by Brazilian agriculture is

advancing. This result is directly related to the increasing deforestation and fires, which have been worsening due to the expansion of agricultural activities, mainly in the Brazilian Amazon [2,3], in addition to the reduction in law enforcement [4]. Another important factor for emissions is the large territorial dimensions occupied by pastures in Brazil and its low animal and plant productivity. Currently, 167.7 million hectares are occupied with pastures, with 77.1% (118.9 million hectares) showing an intermediate or severe degree of degradation [5].

Degraded pastures may present lower short-term greenhouse gas (GHG) emissions than well-managed pastures and integrated systems such as the crop–livestock–forest integration. This is often due to lower soil biological activity in degraded pastures, which can result in less microbial decomposition and, consequently, lower emissions [6]. However, despite the lower short-term GHG emissions, the carbon balance in degraded pastures is often negative due to low or negative soil carbon fixation rates [7]. In this sense, Brazil's Low-Carbon Agriculture Plan (ABC Plan) has incentivized low-carbon agricultural practices by financing, among other measures, the adoption of so-called integrated systems with subsidized credit and, implemented, aiming to offset emissions and improve pasture conditions in Brazil, with ambitious commitments to reduce emissions and recover 15 million hectares of degraded areas [8]. Climate-smart management practices, such as no-tillage, integration systems, biological nitrogen fixation, forest planting, and animal waste treatment, were proposed to make these goals possible [8].

Accordingly, integrated production systems have been implemented across Brazil, including large areas in the Amazon. Integrated systems have a negative net C balance, with the greatest sequestration observed in systems with a forestry component [9]. The implementation of these systems in Brazilian soils has reached significant levels of 11.5 million hectares covered by some type of integration in recent years [10]. These integrated systems are classified as crop–livestock–forest, livestock–forest (LF), crop–livestock, and crop–forest integration systems [11]. Each of these systems plays an important role in enhancing agricultural production, improving animal welfare, and, most importantly, in recovering degraded areas and preserving forest areas [7,12–14].

Specifically, integrated systems with intercropped trees have great potential for increasing C in the soil when compared to other conservation agricultural systems [15,16]. Besides the high amount of litter inputs provided by the natural senescence of leaves, the main tree species introduced in areas of LF in Brazil have deeper and broader root systems, with positive effects on soil C accretion and stabilization [17]. Regarding the overall C balance, it is extremely important to study the effects of these systems on soil attributes, which are drivers of CO₂ emissions [7,10,13]. Many studies have already indicated the influence of soil chemical, physical, and biological attributes on the variation of CO₂ emissions [18–20].

Physical attributes influence the relationships with soil structure and C maintenance in the system, as sudden changes in porosity, bulk density, and aggregation significantly increase C losses in the form of CO₂ [19,20]. Among chemical attributes, organic matter in its different forms and fractions, such as humic substances, stands out by playing an important role in CO₂ mitigation [18,20,21], being the most stable fraction and constituting from 85 to 90% of the total soil organic carbon reserve [21]. However, most of these attributes are related to CO₂ emissions, as they somehow affect the diversity of microorganisms in the soil [22,23]. In addition, humic substances contribute to the growth of microorganisms that help in mineralization and immobilization processes, controlling the dynamics of C in the system [20,22].

Additionally, the season has a significant effect on these biological processes and soil respiration. Season assessments are essential for understanding soil CO₂ emissions in conservationist systems in the Amazon. While the wet season influences soil moisture

and carbon dynamics, the dry season provides insights into system stability and resilience. Including both seasons in evaluations helps develop better strategies for soil health, carbon sequestration, and the long-term sustainability of these systems [9,24]. This indicates that future research should also examine these processes during the dry season.

Given this scenario, the hypothesis of this study is based on the understanding that the conversion of areas under degraded pastures to managed and integrated pasture systems favors lower CO₂ emissions. Thus, knowledge about C balance in integrated production systems is relevant, enabling the search for more resilient and sustainable agricultural systems, capable of contributing to the reduction of GHG emissions. Specifically, this information is even more important for the Amazon region because of its contribution to Brazilian GHG emissions, mainly related to deforestation. Moreover, the Amazon has plenty of degraded pastures, and land recovery is a feasible option for land sparing and avoiding deforestation. Thus, this study aimed to assess the dynamics of CO₂ emissions and their relationship with soil attributes under different Amazon land uses (degraded pastures, managed pastures, and integrated systems).

2. Materials and Methods

2.1. Characterization of the Study Area

The experiment is carried out in the municipality of Paranaíta, located in northern Mato Grosso State, along the MT 206 highway, in the Amazon biome. Its georeferenced position is under the coordinates 9°43'45" S and 56°24'20" W, 250 m above sea level. The region has a mean temperature close to 26.5 °C, and a relative air humidity of 70%.

The climate presents well-defined rainy seasons, with intense precipitation in the summer and drought in the winter, with an annual mean precipitation of around 2300 mm, fitting into the tropical monsoon climate (Am), according to Köppen [25]. The soil of the experimental areas is classified as a dystrophic Red-Yellow Latosol (Oxisol) [26], characteristic soil in the region, with a moderately flat and wavy relief.

The experimental areas were chosen because they are in the Amazon and Cerrado transition (Figure 1), presenting different land use typologies and important characteristics of changes in management, and being close to each other, at 2 km apart, which enables the minimization of the effects of climate variations. The experimental areas were covered by intense native vegetation until the beginning of the 1990s, but the areas were deforested aiming for the transition to the pasture system. The areas were converted to different uses after 2011 and again in 2018 for the treatments in this research. Figure 2 shows the treatments used in this research: degraded pasture (DP) for 10 years without being reformed; managed pasture (MP) for 2 years, with the sowing of *Panicum maximum* cv. Mombasa; livestock–forest integration (LF), composed of interspersed rows with *Eucalyptus* sp. and African mahogany (*Khaya ivorensis*) at a spacing of 5 meters (m) between plants and 15 m between rows; and native forest (NF), used as a baseline for the analyses, as it presented the initial conditions before the land-use changes.

2.2. Delimitation of the Experiment

The experimental area with the different treatments (DP, MP, LF, and NF) was selected from an analysis on the property, which allowed choosing areas distant from borders and roads, to minimize external interference in the research data. A square of 20 × 20 m was measured in each area, totaling 400 m². Twenty-five collection points equidistantly spaced at 5 × 5 m were delimited in each area. Soil CO₂ emissions (FCO₂) were measured in all areas at 25 sampling points in each area. These points were marked with PVC collars (0.10 m in diameter) installed at each point at a depth of 3 cm to couple the chamber from the LI-COR LI-8100 system.

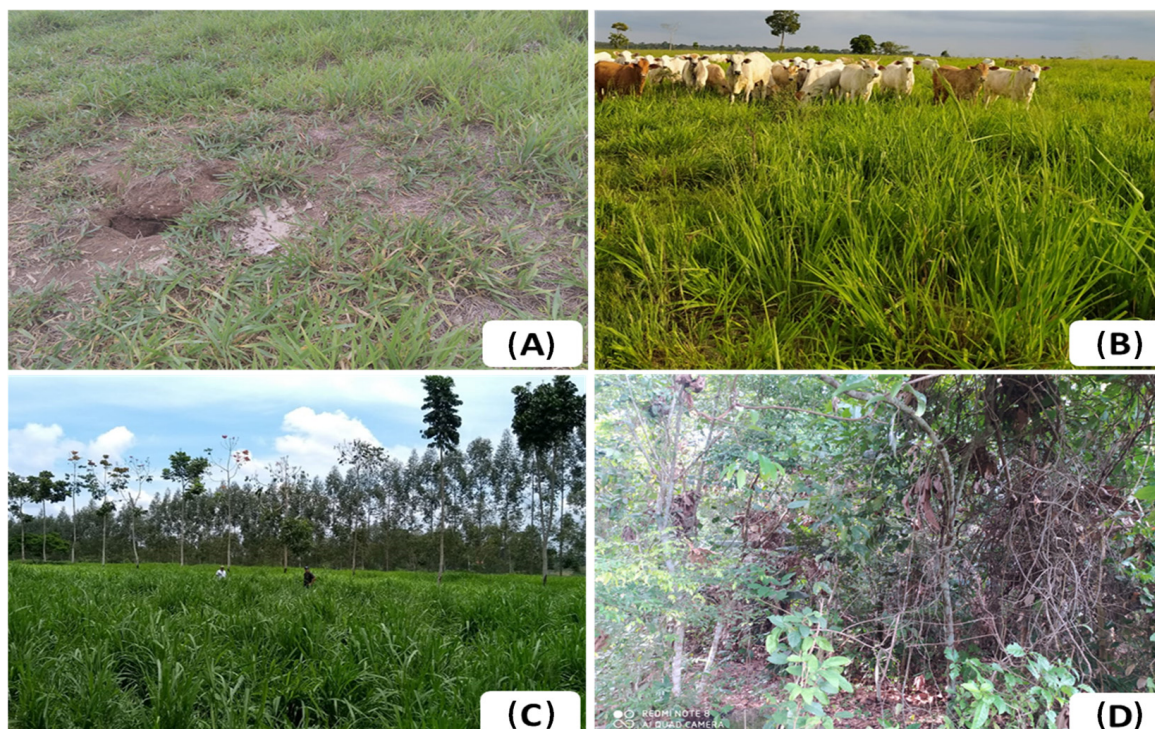


Figure 1. Experimental areas. (A) DP—Degraded Pasture, (B) MP—Managed Pasture, (C) LF—Livestock–Forest Integration, and (D) NF—Native Forest. Paranaíta, MT, Brazil.

2.3. Assessment of Soil CO₂ Emission, Soil Temperature, and Soil Moisture

For the study, 13 assessments were carried out over a total period of 114 days during the rainy season. The assessments began on 14 December 2018, and were completed on 6 April 2019. The monitoring of soil CO₂ emission (FCO₂), soil temperature (Ts), and soil moisture (Ms) were conducted sequentially, with measurements taken every 2 and 3 days in the first week, every 7 days in the second week, and every 15 days in the subsequent weeks until the conclusion of the experiment. FCO₂ was recorded in the morning, from 8:00 to 10:00 a.m. using a portable LI-COR LI-8100 system (LI-COR, Lincoln, NE, USA). This system consists of a closed chamber, which is coupled to collars previously inserted into the soil at the assessment points. This device monitors changes in CO₂ concentration inside the chamber by means of optical absorption spectroscopy in the infrared spectral region (IRGA—Infrared Gas Analyzer, Freiburg im Breisgau, Germany). Ts and Ms were assessed simultaneously with FCO₂ assessments.

Ts was assessed at a depth of 0.15 m using a portable thermometer [27]. Ms was measured using the direct method of Reichardt [28]. Undisturbed soil samples were collected during all assessments to measure the moist mass of the soil to obtain a known value (within a range from 70 to 150 g) for later drying in a forced-air circulation oven at 105 °C for 24 h. Equation (1) was used to calculate this attribute through the difference between the moist and dry soil values:

$$Ms = \frac{M1 - M2}{M2} \quad (1)$$

where M1 is the mass of moist soil freshly collected from the field and M2 is the soil mass after 24 h in an oven at 105 °C.

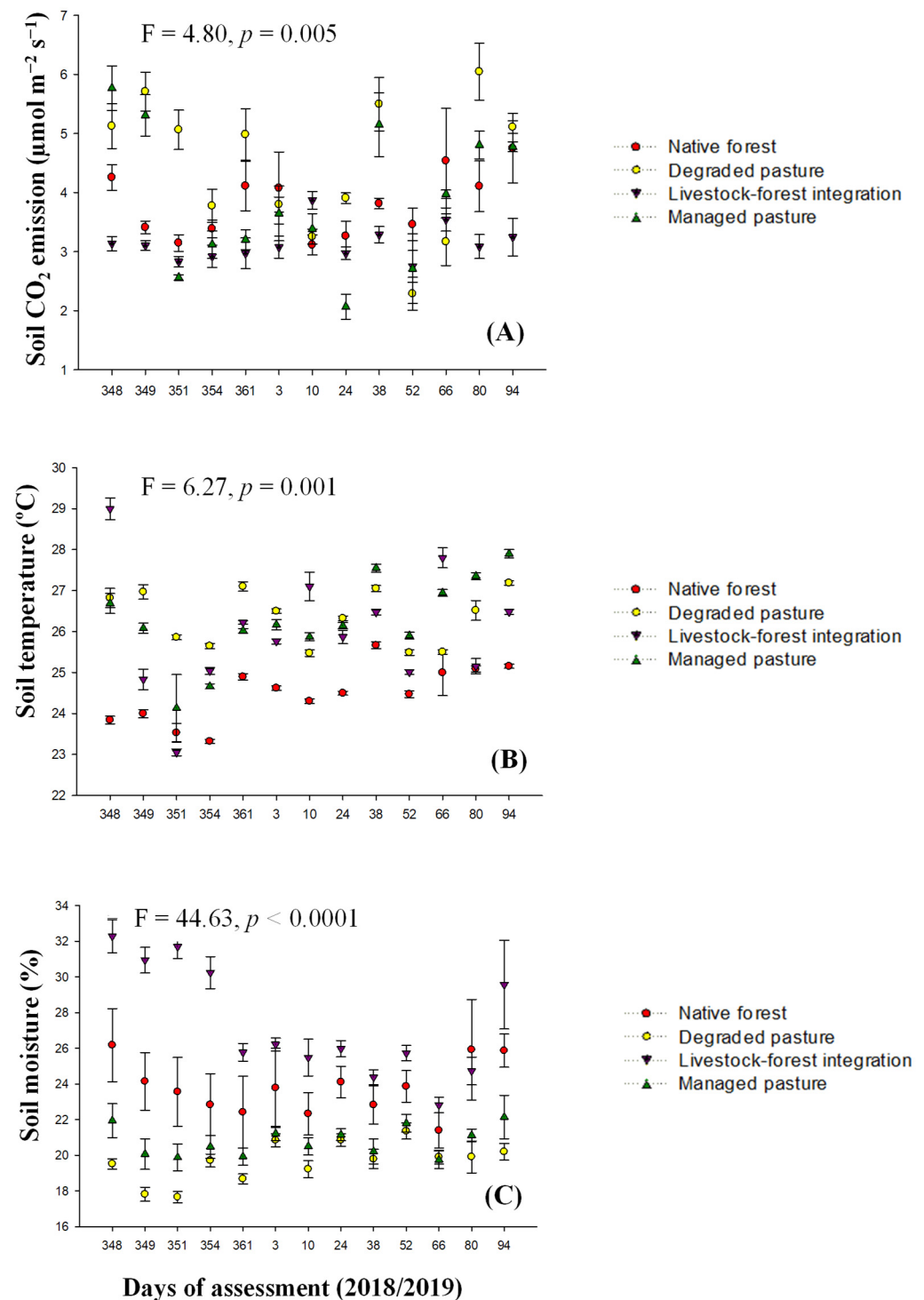


Figure 2. Daily means and mean standard error bars of soil CO₂ emission (A), soil moisture (B), and soil temperature (C) in different land uses, Paranaíta, MT, Brazil, 2018 to 2019.

2.4. Sampling and Soil Physical, Chemical, and Biological Analyses

On the final day of the FCO₂, Ts, and Ms measurements at the conclusion of the experiment (6 April 2019), soil samples were collected for chemical, physical, and biological analysis. Disturbed and undisturbed soil samples were collected from each point in the 0.0–0.20 m depth layer. The samples were taken to the Laboratory of Soil and Leaf Analysis (UNEMAT, Alta Floresta, Brazil), air-dried, homogenized, and sieved on a 2-mm mesh sieve [29] for later use in the analysis.

2.5. Soil Chemical Analyses

Soil pH was determined in H₂O using a benchtop pH meter. The potential acidity (H + Al) was estimated using Ca acetate and determined via titration, while soil texture was determined by the densimeter method (Bouyoucos) [30]. Subsequently, the attributes cation exchange capacity (CEC), soil organic carbon (SOC), and soil carbon stock (Cstock) were calculated.

The attribute SOC was determined using the wet combustion method using colorimetry [31]. Cstock (metric tons ha^{−1}) was assessed by correcting the variation in the areas, calculated by multiplying SOC by soil bulk density (BD measured in grams centimeter^{−3}) and the thickness of the equivalent soil layer (cm), using Equation (2):

$$Cstock = \frac{\left(SOC \cdot BD \cdot \left(\frac{D_{ref}}{D_s} \cdot E \right) \right)}{10} \quad (2)$$

where *Cstock* is the carbon stock (Mg ha^{−1}), *SOC* is the soil organic carbon at the studied depth (grams kilogram^{−1}), *BD* is the soil bulk density at the studied depth, *Dref* is the soil bulk density for the sampled depth in the reference area, and *E* is the studied depth. Finally, the chemical fractionation of soil organic matter was also determined to chemically qualify the contents of complex compounds, such as humic acid (HA), fulvic acid (FA), and humin [31].

2.6. Soil Physical Analyses

Undisturbed soil samples were collected from all 25 sampling points of each area at a depth of 0.0–0.20 m using a sampler adapted to rings of 0.05 m in internal diameter and 0.05 m in height to assess the variability of the treatments. The soil physical aspects macroporosity (Macro), microporosity (Micro), total porosity (TP), soil bulk density (BD), and soil texture (Sand, Silt, and Clay) were characterized. The tension table methodology was used to determine Macro, Micro, and TP. Soil bulk density was also determined [32].

2.7. Soil Biological Analyses

Approximately 300 g of soil was collected using an auger in the 0–0.10-m layer at each sampling point aiming at the soil microbiological characterization. The samples were placed in thermal boxes refrigerated with dry ice until transport to the Laboratory of Soil and Leaf Analysis of UNEMAT to perform the analyses. The samples were pre-packed and passed through a 2-mm mesh sieve before characterization [33]. Subsequently, basal soil respiration (BSR) and soil microbial biomass carbon (MBC) were determined.

2.8. Data Processing and Analysis

2.8.1. Temporal Variation and Descriptive Statistics of Treatments

The data were tabulated and analyzed through repeated-measures analysis for the variables for soil CO₂ emission (FCO₂), soil temperature (Ts), and soil moisture (Ms). Linear regression analyses were performed based on the treatment means on each assessment day to determine the degree of relationship between the analyzed variables. All procedures were performed in the R environment [34]. Descriptive analyses of the results were also performed to better characterize the data, using the mean, standard error, and coefficient of variation for each area studied.

2.8.2. Pearson's Linear Correlation Analysis

The linear relationship of the attributes: soil organic carbon (SOC), soil bulk density (BD), potential acidity (H + Al), Sand, Silt, Clay, soil carbon stock (Cstock), microporosity (Macro), microporosity (Micro), total porosity (TP), potential of hydrogen (pH), cation

exchange capacity (CEC), fulvic acid (FA), humic acid (HA), humin, basal soil respiration (BSR), soil microbial biomass carbon (MBC), Ts, and Ms of each area was studied through the Pearson's linear correlation with the main variable FCO₂. Ts, Ms, and FCO₂ were obtained by averaging the values of the assessment days for each of the 25 replicates in each treatment.

2.8.3. Multivariate Analysis

The principal components whose eigenvalues were higher than the unity were considered in this analysis. The coefficients of linear functions, which define the principal components, were used to interpret their meaning using the signal and relative size of the coefficients as an indication of the weight to be assigned to each variable. Only coefficients with values, usually those higher than or equal to 0.40 in absolute value, were considered in the interpretation. The multivariate analyses were processed using the package gplots of the software R version 4.2.2 [34].

3. Results

3.1. Assessment of Temporal Variability of Soil CO₂ Emission, Soil Temperature, and Soil Moisture

The variables FCO₂, Ts, and Ms showed variations over a period of 114 days, regardless of the treatment (Figure 2A–C). NF and LF presented the lowest mean values of FCO₂ (3.80 and 3.14 $\mu\text{mol m}^{-2} \text{s}^{-1}$, respectively) and the lowest emission ranges throughout the experiment, with variations of 1.63 $\mu\text{mol m}^{-2} \text{s}^{-1}$ for NF and 1.11 $\mu\text{mol m}^{-2} \text{s}^{-1}$ for LF. On the other hand, the highest mean emissions were observed in MP and, primarily, DP, with values of 3.88 and 4.44 $\mu\text{mol m}^{-2} \text{s}^{-1}$, respectively, with the highest emission ranges showing 3.70 $\mu\text{mol m}^{-2} \text{s}^{-1}$ of variation for MP and 3.76 $\mu\text{mol m}^{-2} \text{s}^{-1}$ for DP.

The lowest daily emission values occurred on Julian day 10 for NF (3.11 $\mu\text{mol m}^{-2} \text{s}^{-1}$), 52 for DP (2.28 $\mu\text{mol m}^{-2} \text{s}^{-1}$) and LF (2.75 $\mu\text{mol m}^{-2} \text{s}^{-1}$), and 24 for MP (2.06 $\mu\text{mol m}^{-2} \text{s}^{-1}$). The highest values of CO₂ emission occurred in the assessments carried out on days 94 for NF (4.74 $\mu\text{mol m}^{-2} \text{s}^{-1}$), 80 for DP (6.04 $\mu\text{mol m}^{-2} \text{s}^{-1}$), 10 for LF (3.86 $\mu\text{mol m}^{-2} \text{s}^{-1}$), and 348 for MP (5.76 $\mu\text{mol m}^{-2} \text{s}^{-1}$). These results demonstrate that land-use and management systems are major drivers of CO₂ emissions and their variation during the rainy season in the Amazon.

The lowest Ts values were observed on Julian days 354 for NF (23.32 °C), 10 for DP (25.47 °C), and 351 for LF (23.05 °C) and MP (24.13 °C), while the highest Ts values were observed on days 90 for NF (25.15 °C), DP (27.18 °C), and MP (27.90 °C), and 348 for LF (28.99 °C). Thus, Ts influences on a temporal scale, affecting the highest emission in different land uses, as the highest emissions in NF occurred in the assessment on Julian day 90, which had the peak of maximum temperature. The highest emissions in DP, LF, and MP were observed on days with temperatures close to the maximum values, namely, 80, 10, and 348, respectively. The lowest emissions occurred on days when Ts was closer to the minimum observed in the entire experiment, namely, 10, 52, and 24 for NF, DP, and LF, respectively.

Ms showed less influence on FCO₂ compared to Ts. NF and LF presented higher moisture values, and reductions in Ms values led to an increase in CO₂ emissions. However, little could be explained about the variation in the aspects involving the emissions for these areas. The lowest Ms values were observed on Julian days 66 for NF (21.39%), LF (22.81%), and MP (19.74%), and 351 for DP (17.65%). The highest Ms values were observed on days 348 for NF (26.17%) and LF (32.28%), 3 for DP (20.84%), and 90 for MP (22.14%). Thus, emissions in NF and MP showed no direct relationship with Ms ($p > 0.05$), as both areas presented similar emissions for the highest and lowest Ms values (4.25 and 4.53 $\mu\text{mol m}^{-2} \text{s}^{-1}$; 4.77 and 3.97 $\mu\text{mol m}^{-2} \text{s}^{-1}$, respectively). However, the lowest

emissions in NF were observed on the day of the highest Ms, whereas the lowest emissions in MP were found on the day of the lowest Ms.

The day with the lowest Ms presented the highest FCO₂ value in DP (5.06 $\mu\text{mol m}^{-2} \text{s}^{-1}$), while the highest Ms values were related to low emissions (3.79 $\mu\text{mol m}^{-2} \text{s}^{-1}$), close to the minimum values. The same was found for LF, which presented lower emission values with the highest Ms values (3.13 $\mu\text{mol m}^{-2} \text{s}^{-1}$) and higher emissions when Ms had the lowest values (3.54 $\mu\text{mol m}^{-2} \text{s}^{-1}$). Thus, the highest peaks of CO₂ emission in MP and DP were observed when Ms values were low, mainly in DP, with an inverse relationship between Ms and FCO₂.

As observed, Ms was the attribute that least explained the variation in CO₂ emissions. In contrast, the variable Ts was the attribute that most explained the variation in CO₂ emission in the different areas of study (Figure 3).

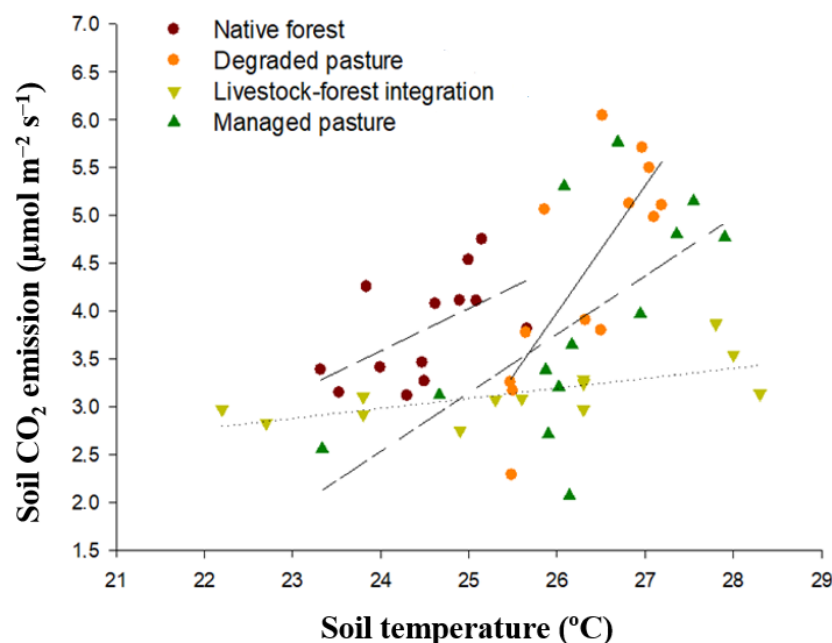


Figure 3. Linear regression between soil CO₂ emission and soil temperature in different land use typologies.

Linear regressions were plotted for all treatments in NF, DP, LF, and MP to assess the distribution of FCO₂ $\times$ Ts. The results show that Ts could explain approximately 32, 60, 47, and 40% of the emission data variability, respectively (Figure 3). It demonstrates the importance of this variable for the regulation of CO₂ emissions on a time scale.

Therefore, Ts acts directly in the regulation of CO₂ dynamics in the different studied areas of the Amazon. Additionally, given the influence of Ts on emissions, the data dispersion showed that NF and LF were the lowest emission sources. This aspect stands out mainly when compared to DP, which was a significant source of CO₂, highlighting the potential of LF to contribute to GHG emission mitigation programs.

3.2. Descriptive Statistics of Soil CO₂ Emissions Associated with Soil Attributes

The lowest mean values of FCO₂ were observed in areas that presented the lowest means of Ms and the highest means of Ts, showing a gradient from higher to lower emissions in DP (FCO₂ = 4.44 $\mu\text{mol m}^{-2} \text{s}^{-1}$; Ts = 26.48 °C; Ms = 19.57%) > MP (FCO₂ = 3.88 $\mu\text{mol m}^{-2} \text{s}^{-1}$; Ts = 26.26 °C; Ms = 21.03%) > NF (FCO₂ = 3.80 $\mu\text{mol m}^{-2} \text{s}^{-1}$; Ts = 24.41 °C; Ms = 22.96%) > LF (FCO₂ = 3.14 $\mu\text{mol m}^{-2} \text{s}^{-1}$; Ts = 26.05 °C; Ms = 27.37%).

The lowest ranges of the mean values of FCO₂ and Ms occurred in LF (0.97; 1.09), MP (2.8; 0.86), and DP (2.41; 1.35), which was reflected in the lowest coefficients of variation, with values corresponding to 8.92, 16.42, and 14.16% for FCO₂ and 4.74, 6.24, and 4.26% for Ms, respectively. In contrast, the lowest ranges for Ts variation were observed in LF (1.09) and MP (0.89), which showed the highest stability in Ts throughout the experimental period (Table 1).

Table 1. Descriptive statistics of soil CO₂ emission, soil temperature, and soil moisture along with soil chemical, physical, and biological attributes in different land uses in the Amazon.

Attribute	Native Forest		Degraded Pasture		Managed Pasture		Livestock–Forest Integration	
	Mean	SE	Mean	SE	Mean	SE	Mean	SE
FCO ₂ (μmol m ^{−2} s ^{−1})	3.80	0.22	4.44	0.13	3.88	0.13	3.14	0.06
Ms (%)	22.96	0.44	19.57	0.17	21.03	0.26	27.37	0.26
Ts (°C)	24.41	0.05	26.48	0.06	26.26	0.05	26.05	0.05
pH (H ₂ O)	4.42	0.05	4.46	0.05	5.33	0.07	4.86	0.03
H + Al (g kg ^{−1})	2.03	0.11	3.04	0.19	2.84	0.14	3.12	0.08
Sand (g kg ^{−1})	701.80	10.9	508.6	8.75	526.8	12.14	561.80	7.21
Silt (g kg ^{−1})	48.95	4.29	43.15	4.19	51.55	6.87	44.95	5.03
Clay (g kg ^{−1})	249.25	10.7	448.25	6.51	421.65	9.55	393.25	5.50
SOC (g kg ^{−1})	4.93	0.33	7.31	0.28	8.20	0.57	6.01	0.36
Cstock (mg ha ^{−1})	12.42	0.83	18.71	0.72	21.49	1.49	16.12	0.97
CEC (cmol _c dm ^{−3})	2.41	0.13	4.07	0.22	4.60	0.17	3.94	0.11
Macro (m ³ m ^{−3})	0.10	0.04	0.05	0.02	0.06	0.02	0.06	0.03
Micro (m ³ m ^{−3})	0.29	0.03	0.28	0.07	0.35	0.03	0.31	0.04
TP (m ³ m ^{−3})	0.39	0.05	0.33	0.07	0.41	0.03	0.36	0.04
BD (g cm ^{−3})	1.27	0.02	1.32	0.01	1.30	0.01	1.35	0.01
FA (g kg ^{−1})	1.42	0.06	1.68	0.07	1.38	0.09	1.77	0.07
HA (g kg ^{−1})	1.13	0.09	0.69	0.12	0.50	0.07	0.38	0.06
Humin (g kg ^{−1})	6.97	0.38	8.31	0.32	9.64	0.36	8.22	0.28
MBC (μg g ^{−1})	152.13	8.84	54.54	3.22	185.13	6.11	108.11	3.01
BSR (mg C- CO ₂ kg ^{−1} soil hour ^{−1})	0.48	0.02	0.51	0.03	0.41	0.02	0.46	0.02

FCO₂: soil CO₂ emission, Ms: soil moisture, Ts: soil temperature, pH: potential of hydrogen, H + Al: potential acidity, SOC: soil organic carbon, Cstock: soil carbon stock, CEC: cation exchange capacity, Macro: macroporosity, Micro: microporosity, TP: total porosity, BD: soil bulk density, FA: fulvic acid, HA: humic acid, MBC: soil microbial biomass carbon, BSR: basal soil respiration. Mean: mean values of the attributes, SE: standard error of the mean.

The chemical attributes pH and H + Al in LF and NF presented the highest stability. These had lower range values, being related to the lowest FCO₂ values in these areas. In addition, NF and LF also had the highest MBC and lowest BSR values and, consequently, the lowest FCO₂ values in the experiment (Table 1). The soil physical attributes Macro, Micro, TP, and BD showed similar mean values between treatments (Table 1). However, NF and LF presented the lowest ranges of variation among the studied treatments, confirming that the lower the range, the higher the homogeneity of physical attributes throughout the studied period and, consequently, the lower the CO₂ emissions in these areas. The lower emissions in NF compared to MP and, mainly, DP, are due to the sandier texture of the area. In this sense, the soil of LF showed a more clayey nature, which may have contributed to its lower emission compared to all other treatments.

Regarding the chemical fractionation of soil organic matter, LF presented the highest FA values (1.77 g kg^{−1}). However, the highest amounts of HA were observed in NF (1.13 g kg^{−1}), while LF had the lowest values (0.38 g kg^{−1}), similar to MP (0.50 g kg^{−1}) and DP (0.69 g kg^{−1}). The highest values for the humin fraction (Hm) were observed in the

managed areas, especially MP (9.64 g kg⁻¹), followed by DP (8.31 g kg⁻¹), LF (8.22 g kg⁻¹), and NF (6.97 g kg⁻¹). Soil organic carbon and carbon stock values were higher in MP (8.20 g kg⁻¹ and 21.49 g kg⁻¹), and LF (6.01 g kg⁻¹ and 16.12 g kg⁻¹), respectively.

The microbiological analyses showed the highest MBC values in MP (185.13 µg g⁻¹) < NF (185.13 µg g⁻¹) < LF (108.11 µg g⁻¹), demonstrating that the areas can maintain MBC. In contrast, the lowest MBC values were found in DP (54.54 µg g⁻¹), which is related to its degradation stage, showing a reduction of MBC.

The lowest BSR values were observed in soils under NF (0.48 mg C- CO₂ kg⁻¹ soil hour⁻¹) and LF (0.46 mg C- CO₂ kg⁻¹ soil hour⁻¹). This higher amount of MBC with a lower BSR may be an indication of lower stress in the system, reflected in the lower CO₂ emissions. DP was the area with the lowest MBC and the highest BSR values (0.51 mg C- CO₂ kg⁻¹ soil hour⁻¹), that is, one of the areas that most contributed to the emissions in the system.

3.3. Pearson's Correlation Between Soil CO₂ Emission and Soil Attributes of the Different Study Areas

Ts was correlated with FCO₂ in all the studied treatments (NF, DP, MP, and LF). It explains the real influence of this attribute, being related to the emission dynamics of this important GHG in the most different studied production systems.

The attributes SOC and BD also had a significant correlation with FCO₂, in addition to Ts, in NF (Table 2). FCO₂ was also correlated with HA in DP. Moreover, FCO₂ was correlated with SOC, H + Al, and MBC in MP and SOC and humin in LF.

Table 2. Correlation analysis between soil CO₂ emission (FCO₂) and soil attributes in the different study areas.

Area	Main Variable		Soil Attribute									
			pH	H + Al	Cstock	SOC	CEC	HA	FA	Hm	Macro	Micro
			(H ₂ O)	(g kg ^{−1})	(Mg ha ^{−1})	(g kg ^{−1})	(cmol _c dm ^{−3})	(g kg ^{−1})	(g kg ^{−1})	(g kg ^{−1})	(%)	(%)
NF	FCO ₂	r	0.12	0.03	0.02	−0.40	0.03	−0.16	0.22	0.06	0.31	−0.31
DP		r	0.30	0.29	−0.25	−0.36	0.10	0.44	0.14	0.12	0.10	0.11
MP		r	−0.28	0.40	−0.17	−0.62	0.06	−0.26	−0.01	−0.02	−0.02	0.03
LF		r	0.01	−0.29	0.00	−0.79	−0.31	−0.31	−0.26	−0.54	−0.19	−0.21
Area	Main variable		TP	BD	MBC	BSR	Sand	Silt	Clay	Ms	Ts	
			(%)	(g.cm ^{−3})	(μg g ^{−1})	(mg C- CO ₂ kg ^{−1} soil hour ^{−1})	(%)	(%)	(%)	(%)	(°C)	
NF	FCO ₂	r	0.06	−0.40	0.16	−0.15	−0.22	0.10	0.18	0.21	0.54	
DP		r	0.13	−0.26	−0.17	0.03	−0.01	0.19	−0.09	−0.29	0.59	
MP		r	0.03	0.07	−0.41	0.15	0.16	0.13	0.11	0.27	0.53	
LF		r	−0.23	−0.22	0.10	0.26	−0.08	−0.01	0.12	−0.28	0.52	

NF: native forest, DP: degraded pasture, MP: managed pasture, and LF: livestock–forest integration. FCO₂: soil CO₂ emission, Ts: soil temperature, Ms: soil moisture. pH: potential of hydrogen, H + Al: potential acidity, SOC: soil organic carbon, Cstock: soil carbon stock, CEC: cation exchange capacity, Macro: macroporosity, Micro: microporosity, TP: total porosity, BD: soil bulk density, FA: fulvic acid, HA: humic acid, MBC: soil microbial biomass carbon, BSR: basal soil respiration. Values in bold: significant correlation by Student's t-test. The values of Ts, Ms, and FCO₂ are the mean of the assessment days for each point in the different treatments, while the values of the other attributes represent a single collection at the end of the experiment.

A negative correlation in NF was observed between BD (−0.407) and SOC (−0.403) and FCO₂. It demonstrates that a reduction in BD and SOC is reflected in increased emissions, explaining approximately 40.7 and 40.3%, respectively.

As the amount of HA increased, DP showed a corresponding rise in FCO₂ (0.44). The same trend was observed for H + Al in MP, where its increment led to an increase in CO₂ emissions (0.40). Negative correlations were observed in MP between SOC and MBC, showing an increase in FCO₂ with a reduction in these attributes (−0.62). Conversely, LF showed a tendency for increased CO₂ emissions in the field with a reduction in Hm (−0.54).

The correlation for Ts was inversely proportional to that observed for SOC for all studied areas, as Ts and FCO₂ showed a positive correlation in NF, DP, MP, and LP (0.54, 0.59, 0.53, and 0.52, respectively), with the correlation coefficient explaining more than 54, 59, 53, and 52% of the effect between both variables, respectively. It confirms that an increase in Ts and a reduction in SOC led to increases in FCO₂ in the systems.

3.4. Interdependence Relationship Between Attributes and Different Amazon Land Uses

In the present study, the two principal components (Dim1 and Dim2) together retained 58% of the original data variation (Figure 4 and Table 3). FCO₂ (0.55) was positively related to Ts (0.87) and BSR (0.51) and negatively related to pH (−0.63), MBC (−0.79), Micro (−0.71), and Us (−0.47). These relationships observed in the biplot in Dim2 indicate that degraded pasture favors soil respiration (FCO₂ and BSR) due to its higher temperature. Furthermore, when observing the confidence ellipses (95% confidence), it is noted that these characteristics are responsible for differentiating the degraded pasture system from the other systems under study, since the ellipses do not intersect.

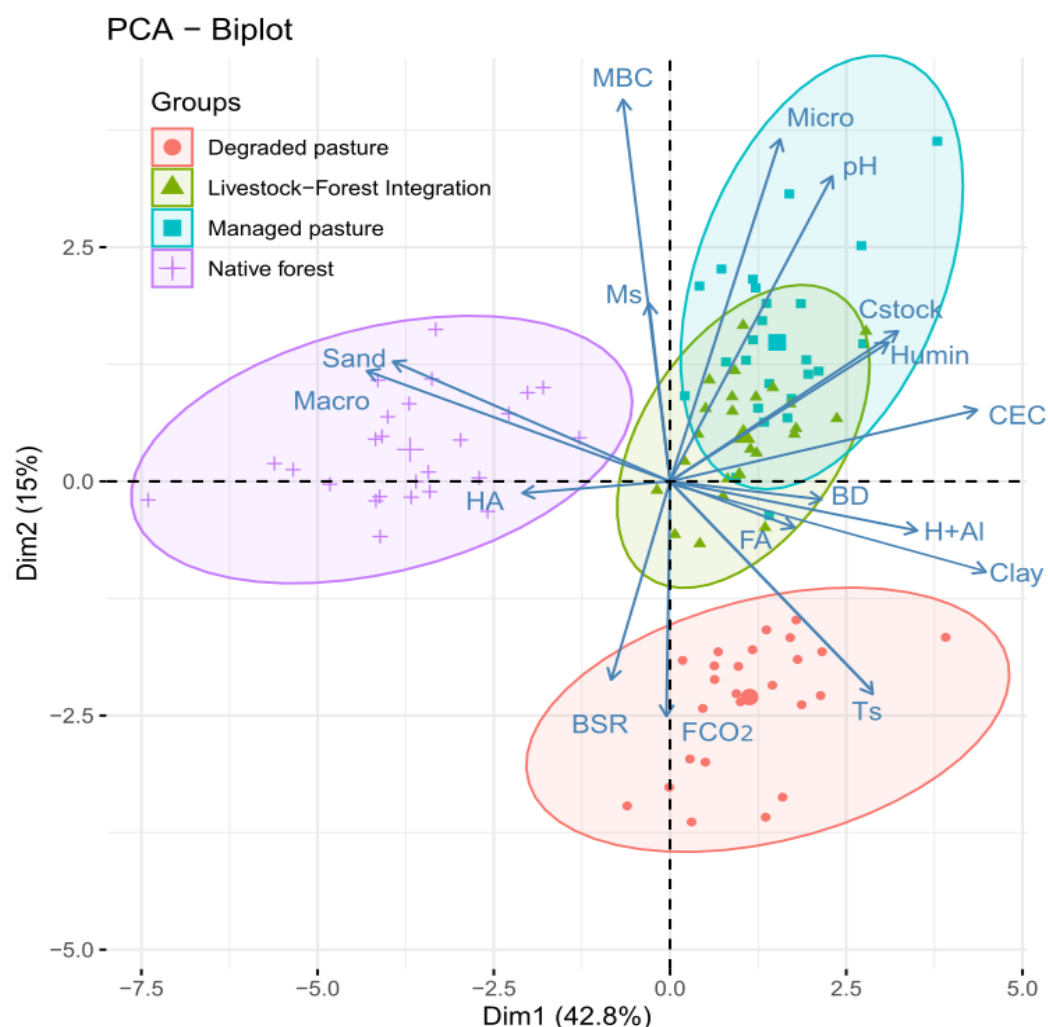


Figure 4. Biplot graph with soil attributes, management systems, and confidence ellipses (95% confidence). FCO₂: soil CO₂ emission, Ts: soil temperature, Ms: soil moisture. pH: potential of hydrogen, H + Al: potential acidity, Cstock: soil carbon stock, CEC: cation exchange capacity, Macro: macroporosity, Micro: microporosity, BD: soil bulk density, FA: fulvic acid, HA: humic acid, MBC: soil microbial biomass carbon, BSR: basal soil respiration.

Table 3. Correlation between FCO₂ and soil attributes and the first two principal components (Dim1-Dim2).

Principal Component	Dim1	Dim2
Variance explained (%)	42.8 *	15 *
FCO ₂	0.01	0.55
Ts	−0.27	0.87
BSR	0.16	0.51
pH	−0.35	− 0.63
MBC	0.13	− 0.79
Micro	−0.30	− 0.71
Us	0.06	− 0.47
Sand	0.83	−0.23
Clay	− 0.65	0.35
Cstock	− 0.63	−0.31
CEC	− 0.84	−0.15
Macro	0.76	−0.25
H + Al	− 0.68	0.10
BD	− 0.42	0.04
FA	−0.34	0.10
HA	0.41	0.02
Humin	− 0.60	−0.29

* Value refers to the percentage of variation of the original dataset retained by the respective factors. Loading values in bold (higher than 0.40 in absolute value) were considered in the component interpretation. FCO₂: soil CO₂ emission, Ts: soil temperature, Ms: soil moisture. pH: potential of hydrogen, H + Al: potential acidity, Cstock: soil carbon stock, CEC: cation exchange capacity, Macro: macroporosity, Micro: microporosity, BD: soil bulk density, FA: fulvic acid, HA: humic acid, MBC: soil microbial biomass carbon, BSR: basal soil respiration.

In addition, other land uses, such as managed pasture and livestock–forest integration, have greater potential for C storage, due to the following characteristics of these soils: pH, microporosity, Clay, CEC, H + Al, BD, and FA, which favor a fraction of the humic substances in the soil that is more stable and resistant to decomposition, humin.

The main characteristic of the native forest that distinguishes it from other land uses is associated with its sandy (sand) and more porous (macropores) structure and the large and complex molecules formed by the decomposition of plant and animal residues, humic acids (HA).

4. Discussion

4.1. Temporal Fluctuations in Soil CO₂ Emissions, Temperature, and Moisture

Soil CO₂ emissions are directly influenced by temperature, moisture, and management practices [35]. Similarly, the main factors that govern the CO₂ dynamics in soils are attributes such as Ts and Ms, which are important for studies that explore the emission dynamics of these GHGs [36].

4.1.1. Soil CO₂ Emission Versus Soil Temperature

Soil CO₂ emission is an activity regulated by soil microorganisms, potentiated by organic matter decomposition and heterotrophic respiration [37]. This microbial activity is regulated by important attributes, especially Ts [37,38]. Thus, understanding the result of temperature increases and the reflection on microbial activity and CO₂ emission is important to assess GHG dynamics in agricultural systems [38].

The forage grown in the area would be one of the explanations for the high Ts values in pasture systems studied. In this case, forage showed no efficiency in the uniform soil cover in the area and soil heating prevention due to its morphological characteristics, and this irregular soil cover confirms the effects of higher temperature and carbon losses in the system [39]. In MP, *Panicum maximum* L. cv. Mombasa exhibits clump development, resulting in irregular clumps. This facilitates the entry of heat into the system, causing soil temperatures to rise (Figure 1B).

The growth characteristic of this forage influenced its spatial distribution and altered the aggregate stability, resulting in lower amounts of organic carbon in the production systems of the Brazilian Savanna [40]. Furthermore, an efficient soil cover is associated with the mildest Ts [41].

The importance of the variation of Ts and emission on a temporal scale could be observed, as the increment in Ts is followed by FCO₂ increases throughout the assessments, regardless of the studied area. Thus, the highest mean emissions were observed in areas that showed similarity in terms of the highest mean temperatures (Figure 2).

Pasture degradation increases CO₂ emissions. This increase is associated with lower soil cover and higher soil temperature (Ts). Higher soil temperatures enhance biological activities, such as carbon mineralization [42], as also observed in DP. The presence of trees in LF improved the micro-environmental conditions in this system and the accumulation of C in tree trunks [9], being associated with cooler Ts [43] and, consequently, lower CO₂ emissions (Figure 1).

4.1.2. Soil CO₂ Emission Versus Soil Moisture

Regarding the lower soil moisture (Ms) values in DP and MP, the absence of trees and less soil cover may be associated with these results. Both areas, mainly DP, present higher levels of soil exposure, resulting in higher temperatures and more water losses by evaporation in both systems, which could be decreased in the presence of trees [44,45].

Previous research has reported the direct effect of Ms on FCO₂ [46,47]. Certainly, the increase in Ms would be an important factor for increases in CO₂ emissions [46,48]. The increase in Ms together with Ts has a direct influence on the activity of decomposing microorganisms, increasing the decomposition rate, and leading to an increase in CO₂ emissions [36].

However, this dependence of Ms with the emission in the field was not observed in this study, possibly because the experiment was carried out during the rainy season and soil moisture was not a limiting factor, as it presented no critical levels to interfere negatively with the soil microbial activity even with Ms oscillations.

4.2. Soil CO₂ Emissions Associated with Soil Attributes

DP presented the highest FCO₂ values when compared to other treatments, which agrees with other authors [7,42,49,50]. In Southern Brazil, degraded pastures release more CO₂ than well-managed systems [7]. Accordingly, degraded pastures showed the highest CO₂ emission than well-managed pastures in China [49].

A possible reason for this widespread tendency of increased CO₂ emission is the lower carbon storage capacity in degraded pastures compared to managed systems [47,50]. In contrast, managed pastures have a higher carbon storage capacity and higher C stocks (Table 1). Despite these agreements, the literature presents some contrasting results when CO₂ fluxes are assessed in degraded pastures [42]. In South Africa, the lowest gross CO₂ emission values were found in moderately degraded pastures when compared to those well-managed, and highly degraded pastures [42].

On the other hand, the lowest mean emission values were observed in LF (Table 1). It shows that LF is an important strategy for mitigating CO₂ emissions in the Amazon region. Integrated systems are a suitable strategy for the sustainable intensification of agriculture in Brazil, increasing food (and bioenergy) production while mitigating global warming by soil C sequestration [51,52].

Moreover, livestock forestry systems are a widely adopted practice for pasture recuperation in Brazil and could contribute to partially recovering soil C stocks in areas of

extensive cattle raising under degradation [43], with positive effects on the chemical and biological indicators of soil quality [52].

The lower emissions in LF were associated with the lowest Ts values, which may have been influenced by soil cover conditions. This is because LF had the presence of a forest component, in addition to well-developed forages. Accordingly, NF presented the arboreal component, which also contributed to the thermal balance of the environment. Thus, the lowest temperatures were observed in these two areas studied (Table 1). Degraded pastures are expected to show higher Ts than managed pastures [7], as observed in the present study.

Regarding the soil pH and H + Al in the studied treatments, NF and DP exhibited the highest levels of soil acidity. In contrast, MP had the lowest acidity values and, consequently, the highest CEC. This effect may indicate that reform practices are important for the system, as intensive management contributed to the buffering effect of soil pH and resulted in the lowest CO₂ emissions compared to DP (Table 1).

Soil physical attributes exhibited varying values among areas, with MP treatment showing the highest Micro and TP values, while NF had the highest values of Macro, not differing from MP, and the lowest BD values. It shows a little homogeneity between treatments. Attributes such as porosity can contribute to the variation of CO₂ emissions, with a positive relationship of porosity in different soils with similar management systems [53]. In general, soils with higher total porosity tend to present higher emissions, as it affects microbial activity, microbial biomass and organic matter content [54]. However, the variation of soil porosity in dark earth sites under cacao cultivation can influence the increase or reduction of CO₂ emissions [55].

The highest HA and FA values in LF and NF, respectively, can be attributed to their potentially more dynamic systems and the presence of a forest component. This contributes to higher amounts of these complex organic matter fractions, which are related to an increase in litter on the soil surface. Humins are an important organic matter fraction, with a more lignified constitution, being more difficult to be metabolized by the microbiota. Thus, the amount of this fraction in MP is due to an increase in the degradation of the other acidic fractions by microorganisms [56], with the most complex fraction of organic matter standing out.

Anthropized systems such as MP and LF with higher C stock values than NF have been observed in other studies, such as no-tillage [57], agroforestry [58,59], and silvopastoral integration systems [60]. These management systems can be win-win strategies for climate change mitigation due to soil C accumulation, avoiding GHG emissions. In this sense, these results reinforce that LF and MP are potential systems for use, aiming at effective strategies to prevent CO₂ emissions and improve soil quality, besides the forest and forage components, and have the potential to sequester carbon by fixing CO₂ in the biomass, mainly in the trees of the system [61].

MBC is considered the portion of the soil that involves the entire active living fraction of microorganisms, which are organic matter transforming agents [62]. Accordingly, the highest MBC values were found in treatments that have the highest contents of organic matter and, consequently, organic carbon (Table 1, MP < NF < LF < DP). Similarly, Abdalla et al. [42] observed lower MBC values in DP when assessing pastures with different degradation levels, with values of 55 and 81% higher in non-degraded pastures compared to moderately and highly degraded pastures, respectively. Thus, this increase can contribute to higher control over nutrient availability [62] and microbial growth, reducing BSR [63].

Ghorbani et al. [64] in researching CO₂ emissions affected by land use in different humid areas, observed that pasture and forest areas increased soil organic carbon and MBC compared to other land uses (rice land, kiwi orchard, tea land, and uncultivated land).

4.3. Correlations Between Soil CO₂ Emission and Soil Attributes

The correlations found in this study confirm the important influences of soil attributes on FCO₂ dynamics in different land uses and management systems (Figure 4, Tables 2 and 3). Soil CO₂ emissions are the result of a complex interaction between soil chemical, physical, and biological attributes [21,54,57], also being the major emission regulators in time and space. The influence of Ts on CO₂ emissions is related to its influence on microbial activity, since the ideal values of this attribute followed by increases in Ts can act to potentiate increases in CO₂ emissions due to a higher activity of the microbiota on soil organic matter decomposition [60,65].

The highest Ts value was found in DP, which provided the highest FCO₂ values, confirming what Silva et al. [66] observed when evaluating CO₂ emission in different seasons. The rainy season showed the highest emissions in the pasture system, always at the highest Ts, highlighting the influence of this attribute and season on the emission dynamics. However, through multivariate analysis, it was observed that FCO₂ also presents an inverse relationship with soil moisture (Figure 4 and Table 3).

The relationship between soil moisture and CO₂ emissions is complex. It is known that soil water content can influence FCO₂ in different ways, both inhibiting and optimizing soil respiration [18–20]. Maier et al. [67] when assessing the CO₂ flux and the influences of attributes in the Amazon Forest, the authors stated that Ms is the main attribute controlling emissions in forests [67], while Ts was the main driver of CO₂ emissions in our assessment (Figures 3 and 4).

In their study in tropical dry forests, Vargas-Terminel et al. [68] observed that soil respiration decreased in accordance with soil moisture availability. The authors discussed that understanding the intra-annual and interannual variability of soil respiration and the mechanisms underlying this crucial process remains poorly explored in water-limited ecosystems, and that seasonality plays a critical role in determining the temporal patterns and interactions between biotic and abiotic drivers.

In addition, important relationships in emission reductions and increases have been found with other attributes, such as BD [48,54,69], organic matter [20], Ms and porosity [57], and carbon stock [20,54], which are some of the attributes that showed important relationships with FCO₂ in the present study. A negative correlation was observed between BD and CO₂ emissions (Table 2). CO₂ emissions were inversely proportional to the ratio of humification $\times$ BD in sugarcane areas with burning, with increases in CO₂ emissions following a reduction in BD [48], with similar results also observed for sugarcane systems without burning (green cane) [18].

The less-dense soil in this study has higher pore space, facilitating oxygen entry into the pores. This contributes to higher rates of soil organic matter decomposition and CO₂ release, especially in soils with lower organic matter content [70].

Soil organic matter losses led to an increase in CO₂ emission rates. Possibly, this inverse effect clarifies the importance of SOC in maintaining FCO₂ to avoid losses. Thus, the higher emission values may be a consequence of the low amounts of organic matter, which is composed of approximately 58% C in its structure [71].

The reduction in SOC was possibly a reflection of the CO₂ loss from the system, demonstrating its potential to be influenced by these attributes. Therefore, increasing SOC content may be one of the strategies for more sustainable production systems [72]. Thus, the search for viable strategies to maintain and increase SOC, such as no-tillage [73–75], and integrated livestock systems [7,76], is essential. As observed in the present study (Figure 4), managed pasture and livestock–forest integration systems have greater potential to store C in the soil (Figure 4).

Regarding the chemical fractionation of organic matter, the inverse relationship between Hm and FCO₂ is explained by the fact that this fraction is resistant to biodegradation because it is more recalcitrant and physically protected by clay-humic complexes [77]. Thus, soil microorganisms will preferentially decompose the more biochemically labile fractions of soil organic matter, such as humic and fulvic acids, which require lower energy costs [41]. However, microorganisms may also utilize even the most resistant soil organic fractions, such as Hm, especially in environments with high microbial activity and limited C inputs [56].

The reduction in MBC has a direct consequence of a possible reduction in the microbial activity in the soil. Microbial biomass is the most degradable and bioactive fraction of soil organic matter [78], and pasture management can influence the size and activity of microbial biomass [79]. Moitinho et al. [54] observed a relationship between MBC and CO₂ emission, in which areas with the highest values of this attribute presented the highest emissions.

MBC is closely related to soil carbon and, therefore, the higher the biomass content, the higher the potential for transforming soil organic matter into more stable compartments, thus, reflecting a higher C maintenance in the environment. The increase of organic matter in the system is an important soil carbon maintenance practice because the system becomes more vulnerable and unbalanced as MBC is reduced, resulting in higher CO₂ emissions.

5. Conclusions

The data presented in this locally conducted study during the rainy season demonstrated that the variability of soil CO₂ emissions in the Amazon is influenced by different land uses and is directly affected by soil temperature across all treatments. The integration livestock–forest system results in lower CO₂ emissions, followed by native forest, while managed and degraded pastures have the highest emissions due to lower soil moisture and higher soil temperature. Despite higher carbon stock in managed pastures, the integration livestock–forest system is more effective at stabilizing carbon, maintaining similar carbon stock levels with less CO₂ loss.

Correlations were found between other soil attributes in the different land uses, explaining the variations in the dynamics of CO₂ emissions. In addition to the direct relationship between CO₂ emissions and soil temperature for all treatments, the amount of humic acid induced the highest emission in the degraded pasture. The increase in H + Al contents and the reduction in microbial biomass carbon increased emissions in the managed pasture.

The maintenance of degraded pastures contributes to increased CO₂ emissions. Thus, the recovery of degraded pastures and conversion into the integration livestock–forest system contributes to reducing soil CO₂ emissions in the Amazon soil.

Given that soil respiration is influenced by seasonality, future studies should further investigate the behavior of CO₂ emissions in different Amazon soil land uses during both the dry and rainy seasons.

Author Contributions: Conceptualization, A.M.d.R., A.R.P. and N.L.S.J.; methodology, A.M.d.R., A.R.P. and N.L.S.J.; software, A.M.d.R., A.R.P. and N.L.S.J.; validation formal analysis, A.M.d.R., M.F., A.R.P., O.M.Y. and N.L.S.J.; investigation, A.M.d.R., A.R.P. and N.L.S.J.; resources, A.M.d.R., A.R.P. and N.L.S.J.; data curation, A.R.P., M.R.M., M.A.C.d.C., M.V.M.F. and N.L.S.J.; writing—original draft preparation, A.M.d.R., M.F., M.R.M., M.V.M.F. and N.L.S.J.; writing—review and editing, A.M.d.R., M.F., M.R.M., M.V.M.F., D.M.d.S.O., D.A.F.d.F., O.M.Y. and N.L.S.J.; visualization, A.M.d.R., M.F., M.R.M., M.V.M.F., D.M.d.S.O., D.A.F.d.F., O.M.Y. and N.L.S.J.; supervision, A.R.P. and N.L.S.J.; project administration, A.M.d.R., A.R.P. and N.L.S.J.; design of the work, A.M.d.R., M.F., A.R.P., M.A.C.d.C., M.R.M., M.V.M.F., O.M.Y. and N.L.S.J.; drafting the work, A.M.d.R., M.F., A.R.P., M.A.C.d.C., M.R.M., M.V.M.F., D.M.d.S.O., D.A.F.d.F., O.M.Y. and N.L.S.J.; revising it critically, A.M.d.R., M.F., A.R.P.,

M.A.C.d.C., M.R.M., M.V.M.F., D.M.d.S.O., D.A.F.d.F., O.M.Y. and N.L.S.J. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Data Availability Statement: The data that support the findings of this study are available from the corresponding author, [N.L.S.J.], on reasonable request.

Acknowledgments: The authors would like to thank the Mato Grosso State University (UNEMAT) and the School of Agricultural and Veterinary Sciences of the São Paulo State University (FCAV–UNESP). D.M.S.O is grateful for the support of the RCGI—Research Centre for Gas Innovation, hosted by the University of São Paulo (USP) and sponsored by FAPESP—São Paulo Research Foundation (2014/50279-4 and 2020/15230-5).

Conflicts of Interest: The authors declare no conflict of interest.

References

1. Análise: Quais Países São Historicamente Responsáveis pelas Mudanças Climáticas? Available online: <https://www.carbonbrief.org/analysis-which-countries-are-historically-responsible-for-climate-change> (accessed on 25 November 2021).
2. Painel Intergovernamental Sobre Mudanças Climáticas. *Climate Change 2014: Synthesis Report. Contribution of Working Groups I, II and III to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*; Pachauri, R.K., Meyer, L.A., Eds.; IPCC: Geneva, Switzerland, 2014; p. 151.
3. Brito, M.R.; Siquiera, F.L.T.; De Sousa, I.J.A.; Sousa, R. Estoque de carbono no solo sob diferentes condições de cerrado. *Desafios Rev. Interdiscip. Univ. Fed. Tocantins* **2018**, *5*, 114–124. Available online: https://www.researchgate.net/publication/335468690_ESTOQUE_DE_CARBONO_NO_SOLO_SOB_DIFERENTES_CONDICOES_DE_CERRADO#fullTextFileContent (accessed on 11 December 2021). [CrossRef]
4. Gatti, L.V.; Cunha, C.L.; Marani, L.; Cassol, H.L.G.; Messias, C.G.; Arai, E.; Denning, A.S.; Soler, L.S.; Almeida, C.; Setzer, A.; et al. Increased Amazon carbon emissions mainly from decline in law enforcement. *Nature* **2023**, *621*, 318–323. [CrossRef]
5. Atlas Digital das Pastagens Brasileiras. Available online: <https://atlasdaspastagens.ufg.br> (accessed on 15 December 2021).
6. Meier, E.A.; Thorburn, P.J.; Bell, L.W.; Harrison, M.T.; Biggs, J.S. Greenhouse Gas Emissions from Cropping and Grazed Pastures Are Similar: A Simulation Analysis in Australia. *Front. Sustain. Food Syst.* **2020**, *3*, 121. [CrossRef]
7. Figueiredo, E.B.; Panosso, A.R.; Bordonal, R.O.; Teixeira, E.B.; Berchielli, T.T.; La Scala, N., Jr. Soil CO₂–C Emissions and Correlations with Soil Properties in Degraded and Managed Pastures in Southern Brazil. *Land Degrad. Dev.* **2016**, *28*, 1263–1273. [CrossRef]
8. Brasil Ministério da Agricultura, Pecuária e Abastecimento. *Plano Setorial de Mitigação e de Adaptação às Mudanças Climáticas para a Consolidação de Uma Economia de Baixa Emissão de Carbono na Agricultura: Plano ABC*; MAPA/ACS: Brasília, Brazil, 2012; pp. 1–173.
9. Monteiro, A.; Barreto-Mendes, L.; Fanchone, A.; Morgavi, D.P.; Pedreira, B.C.; Magalhães, C.A.S.; Abdalla, A.L.; Eugène, M. Crop-livestock-forestry systems as a strategy for mitigating greenhouse gas emissions and enhancing the sustainability of forage-based livestock systems in the Amazon biome. *Sci. Total Environ.* **2024**, *906*, 167396. [CrossRef] [PubMed]
10. Lopes, L.B.; Pitta, R.M.; Eckstein, C.; Pedreira, B.C.; Grossi, P.C.; Sindeaux, E.; Peruffo, R.G.; Cornelissen, T.G. Diversity of coleopterans associated with cattle dung in open pastures and silvopastoral systems in the Brazilian Amazon. *Agrofor. Syst.* **2020**, *94*, 2277–2287. [CrossRef]
11. Gil, J.; Siebold, M.; Berger, T. Adoption and development of integrated crop–livestock–forestry systems in Mato Grosso, Brazil. *Agric. Ecosyst. Environ.* **2015**, *199*, 394–406. [CrossRef]
12. Costa, M.P.; Schoeneboom, J.C.; Oliveira, S.A.; Vinas, R.S.; de Medeiros, G.A. A socio-economic efficiency analysis of integrated and non-integrated crop-livestock-forestry systems in the Brazilian Cerrado based on LCA. *J. Clean. Prod.* **2018**, *171*, 1460–1471. [CrossRef]
13. De Souza, K.W.; Pulrolnik, K.; Júnior, R.G.; Marchão, R.L.; Vilela, L.; de Carvalho, A.M.; de Oliveira, A.D. *Offsetting Greenhouse Gas (GHG) Emissions Through Crop-Livestock-Forest Integration*; Embrapa Cerrados: Brasília, Brazil, 2020; pp. 1–13.
14. Dos Reis, J.C.; Kamo, M.Y.T.; Michetti, M.; Wruck, F.J.; Rodrigues, R.A.R.; Farias Neto, A.L. Economic and environmental impacts of integrated systems adoption in Brazilian agriculture-forest frontier. *Agroforest Syst.* **2023**, *97*, 847–863. [CrossRef]
15. Shi, L.; Feng, W.; Xu, J.; Kuzyakov, Y. Agroforestry systems: Meta-analysis of soil carbon stocks, sequestration processes, and future potentials. *Land. Degrad. Dev.* **2018**, *29*, 3886–3897. [CrossRef]
16. Waheed, R.; Chang, D.; Sarwar, S.; Chen, W. Forest, agriculture, renewable energy, and CO₂ emission. *J. Clean. Prod.* **2018**, *172*, 4231–4238. [CrossRef]
17. Zinn, Y.L.; Lal, R.; Resck, D.V.S. Eucalypt plantation effects on organic carbon and aggregation of three different-textured soils in Brazil. *Soil Res.* **2011**, *49*, 614–624. [CrossRef]

18. Moitinho, M.R.; Ferraudo, A.S.; Panosso, A.R.; Bicalho, E.S.; Teixeira, D.B.; Barbosa, M.A.; Tsai, S.M.; Borges, B.M.F.; Cannavan, F.S.; Souza, J.A.M.; et al. Effects of burned and unburned sugarcane harvesting systems on soil CO₂ emission and soil physical, chemical, and microbiological attributes. *Catena* **2021**, *196*, 104903. [\[CrossRef\]](#)
19. Chen, X.; Dhungel, J.; Bhattarai, S.P.; Torabi, M.; Pendergast, L.; Midmore, D.J. Impact of oxygation on soil respiration, yield and water use efficiency of three crop species. *Plant Ecol.* **2011**, *4*, 236–248. [\[CrossRef\]](#)
20. Santos, G.A.A.; Moitinho, M.R.; Oliveira, B.S.; Xavier, C.V.; Teixeira, D.B.; Corá, J.E.; La Scala, N., Jr. Effects of long-term no-tillage systems with different succession cropping strategies on the variation of soil CO₂ emission. *Sci. Total Environ.* **2019**, *686*, 413–424. [\[CrossRef\]](#) [\[PubMed\]](#)
21. Lal, R. Challenges and opportunities in soil organic matter research. *Eur. J. Soil Sci.* **2009**, *60*, 158–169. [\[CrossRef\]](#)
22. Luo, J.; Song, Y.; Liang, J.; Li, I.; Li, T. Elevated CO₂ mitigates the negative effect of CeO₂ and Cr₂O₃ nanoparticles on soil bacterial communities by alteration of microbial carbon use. *Environ. Pollut.* **2020**, *263*, 1–11. [\[CrossRef\]](#)
23. Tulio, R.H.; Rachwal, M.F.G.; Zanatta, J.A.; da Silva, K.; Kaschuk, G. Physical, chemical and microbiological soil attributes influence soil greenhouse gases fluxes in Atlantic Forest and pine (*Pinus taeda*) plantations in Brazil. *Soil Use Manag.* **2023**, *39*, 183–197. [\[CrossRef\]](#)
24. Levine, P.A.; Randerson, J.T.; Chen, Y.; Pritchard, M.S.; Xu, M.; Hoffman, F.M. Soil moisture variability intensifies and prolongs eastern Amazon temperature and carbon cycle response to El Niño-Southern Oscillation. *J. Clim.* **2019**, *32*, 1273–1292. [\[CrossRef\]](#)
25. Alvares, C.A.; Stape, J.L.; Sentelhas, P.C.; Gonçalves, J.L.M.; Sparovek, G. Köppen's climate classification map for Brazil. *Meteorol. Z.* **2013**, *22*, 711–728. [\[CrossRef\]](#)
26. Santos, H.G.; Jacnime, P.K.T.; Anjos, L.H.C.; Oliveira, V.A.; Lumberras, J.F.; Coelho, M.R.; Almeida, J.A.; Araujo Filho, J.C.; Oliveira, J.B.; Cunha, T.J.F. *Sistema Brasileiro de Classificação de Solos*, 5th ed.; Embrapa: Brasília, Brazil, 2018; p. 356.
27. Graf, A.; Weihermüller, L.; Huisman, J.A.; Herbst, M.; Bauer, J.; Vereecken, H. Measurement depth effects on the apparent temperature sensitivity of soil respiration in field studies. *Biogeosciences* **2008**, *5*, 1175–1188. [\[CrossRef\]](#)
28. Reichardt, K. *Processos de Transferência no Sistema Solo-Planta-Atmosfera*, 4th ed.; Fundação Cargill: Campinas, Brazil, 1985; p. 466.
29. *Manual de Análises Químicas de Solos, Plantas e Fertilizantes*, 2nd ed.; Silva, F.C., Ed.; Embrapa: Brasília, Brazil, 2009; p. 356.
30. Empresa Brasileira de Pesquisa Agropecuária. *Manual de Métodos de Análises de Solo*, 2nd ed.; Embrapa: Brasília, Brazil, 1997; p. 212.
31. Empresa Brasileira de Pesquisa Agropecuária. *Manual de Métodos de Análises de Solo*, 3rd ed.; Embrapa: Brasília, Brazil, 2017; p. 577.
32. Empresa Brasileira de Pesquisa Agropecuária. *Manual de Métodos de Análises de Solo*, 2nd ed.; Embrapa: Brasília, Brazil, 2011; p. 225.
33. Silva, E.E.; Azevedo, P.H.S.; De-poli, H. *Determinação da Respiração Basal do Solo (RBS) e Quociente Metabólico do Solo (q CO₂)*; Embrapa: Seropédica, Rio de Janeiro, Brazil, 2007; p. 4.
34. R Core Team. *R: Uma Linguagem e Ambiente para Computação Estatística*; R Foundation for Statistical Computing: Vienna, Austria, 2023.
35. Yuste, J.C.; Baldocchi, D.B.; Gershenson, A.; Goldstein, A.; Misson, L.; Wong, S. Microbial soil respiration and its dependency on carbon inputs, soil temperature and moisture. *Glob. Chang. Biol.* **2007**, *13*, 2018–2035. [\[CrossRef\]](#)
36. Smith, K.A.; Ball, T.; Conen, F.; Dobbie, K.E.; Massheder, J.; Rey, A. Exchange of greenhouse gases between soil and atmosphere: Interactions of soil physical factors and biological processes. *Eur. J. Oral Sci.* **2018**, *69*, 10–20. [\[CrossRef\]](#)
37. Bond-Lamberty, B.; Thomson, A. Temperature-associated increases in the global soil respiration record. *Nature* **2010**, *464*, 579–582. [\[CrossRef\]](#)
38. Chevallier, T.; Hmaid, K.; Kouakoua, E.; Bernoux, M.; Gallali, T.; Toucet, J.; Barthès, B.G. Physical protection of soil carbon in macroaggregates does not reduce the temperature dependence of soil CO₂ emissions. *J. Soil Sci. Plant Nutr.* **2015**, *178*, 592–600. [\[CrossRef\]](#)
39. Thomazini, A.; Mendonça, E.S.; Souza, J.L.; Cardoso, I.M.; Garbin, M.L. Impact of organic no-till vegetables systems on soil organic matter in the Atlantic Forest biome. *Sci. Hortic.* **2015**, *182*, 145–155. [\[CrossRef\]](#)
40. Borges, C.S.; Ribeiro, B.T.; Wendling, B.; Cabral, D.A. Soil aggregation, organic carbon and CO₂ emission in different land uses in Brazilian Savanna, Triangulo Mineiro region. *Amb. Água* **2015**, *10*, 660–675. [\[CrossRef\]](#)
41. Da Silva, D.M.N.; Venturim, C.H.P.; Valory, M.E.O.; de Oliveira, F.L.; de Sá Mendonça, E. Impact of soil cover systems on soil quality and organic production of yacon. *Sci. Hortic.* **2018**, *235*, 407–412. [\[CrossRef\]](#)
42. Abdalla, K.; Mutema, M.; Chivenge, P.; Everson, C.; Vicent, C. Grassland degradation significantly enhances soil CO₂ emission. *Catena* **2018**, *167*, 284–292. [\[CrossRef\]](#)
43. Alves, B.J.R.; Madari, B.E.; Boddey, R.M. Integrated crop–livestock–forestry systems: Prospects for a sustainable agricultural intensification. *Nutr. Cycl. Agroecosyst.* **2017**, *108*, 1–4. [\[CrossRef\]](#)
44. Glatzle, S.; Stuerz, S.; Giese, M.; Pereira, M.; de Almeida, R.G.; Bungenstab, D.J.; Macedo, M.C.M.; Asch, F. Seasonal Dynamics of Soil Moisture in an Integrated-Crop-Livestock-Forestry System in Central-West Brazil. *Agriculture* **2021**, *11*, 245. [\[CrossRef\]](#)

45. Villanueva-López, G.; Martínez-Zurimendi, P.; Ramírez-Avilés, L.; Aryal, D.R.; Casanova-Lugo, F. Live fences reduce the diurnal and seasonal fluctuations of soil CO₂ emissions in livestock systems. *Agron. Sustain. Dev.* **2016**, *36*, 1–9. [CrossRef]
46. Oliveira, C.F. Variabilidade Espacial da Emissão de CO₂ e Estoque de Carbono do Solo em Áreas de Eucalipto e Sistema Silvopastoril. Master's Thesis, Faculdade de Engenharia de Ilha Solteira, Universidade Estadual Paulista Júlio de Mesquita Filho, Ilha Solteira, Brazil, 2018.
47. De Oliveira, D.C.; Maia, S.M.F.; Freitas, R.C.A.; Cerri, C.E.P. Changes in soil carbon and soil carbon sequestration potential under different types of pasture management in Brazil. *Reg. Environ. Change* **2022**, *22*, 87. [CrossRef]
48. Panosso, A.R.; Marques, J., Jr.; Pereira, G.T.; La Scala, N., Jr. Spatial and temporal variability of soil CO₂ emission in a sugarcane area under green and burn managements. *Soil Till. Res.* **2009**, *105*, 275–282. [CrossRef]
49. Chen, J.; Zhou, X.; Wang, J.; Hrusca, T.; Shi, W.; Cao, J.; Zhang, B.; Xu, G.; Chen, Y.; Luo, Y. Grazing exclusion reduced soil respiration but increased its temperature sensitivity in a Meadow Grassland on the Tibetan Plateau. *Methods Ecol. Evol.* **2016**, *6*, 675–687. [CrossRef]
50. Li, Y.; Dong, S.; Liu, S.; Zhou, H.; Gao, Q.; Cao, G.; Wang, X.; Su, X.; Zhang, Y.; Tang, L.; et al. Seasonal changes of CO₂, CH₄ and N₂O fluxes in different types of alpine grassland in the Qinghai-Tibetan Plateau of China. *Soil Biol. Biochem.* **2015**, *80*, 306–314. [CrossRef]
51. Brasil Ministério da Agricultura, Pecuária e Abastecimento—MAPA. Plano Setorial para Adaptação à Mudança do Clima e Baixa Emissão de Carbono na Agropecuária 2020–2030. Available online: <https://www.gov.br/agricultura/pt-br/assuntos/sustentabilidade/plano-abc/arquivopublicacoes-plano-abc/final-isbn-plano-setorial-para-adaptacao-a-mudanca-do-clima-e-baixa-emissao-de-carbono-na-agropecuaria-compactado.pdf> (accessed on 15 December 2021).
52. Oliveira, D.M.S.; Santos, R.S.; Chizzotti, F.H.M.; Bretas, I.L.; Franco, A.L.C.; Lima, R.P.; Freitas, D.A.F.; Cherubin, M.R.; Cerri, C.E.P. Crop, livestock, and forestry integration to reconcile soil health, food production, and climate change mitigation in the Brazilian Cerrado: A review. *Geoderma Reg.* **2024**, *37*, e00796. [CrossRef]
53. Mantovanelli, B.C.; Coutinho Alho, L.; Campos, M.C.C.; Cunha, J.M.; Amorim Oliveira, L. Pedoindicators attributes in the variation of CO₂ efflux in Indian black earth and non-anthropogenic soils. *Carbon Manag.* **2020**, *11*, 1–13. [CrossRef]
54. Moitinho, M.R.; Teixeira, D.; Bicalho, E.; Panosso, A.R.; Ferraud, A.S.; Pereira, G.T.; Tsai, S.M.; Borges, B.M.F.; La Scala, N., Jr. Soil CO₂ emission and soil attributes associated with the microbiota of a sugarcane area in southern Brazil. *Sci. Rep.* **2021**, *11*, 1–15. [CrossRef]
55. Silva, D.M.P.; Campos, M.C.C.; Franciscan, U.; Alho, L.C.; Santos, L.A.C.; Neto, P.P.; Bergamin, A.C.; Souza, Z.M. Spatial variability of soil properties in archeological dark earth sites under cacao cultivation. *Rev. Bras. Ciênc. Solo* **2016**, *40*, 1–12. [CrossRef]
56. Barros, J.D.S.; Chaves, L.H.G.; Chaves, I.B.; Pereira, W.E. Effects of Different Soil Management Systems in the Chemical Properties in the Coastal Plains of State Paraíba. Iran. *J. Energy Environ.* **2011**, *2*, 339–347. [CrossRef]
57. Xavier, C.V.; Moitinho, M.R.; De Bortoli Teixeira, D.; Santos, G.A.A.; de Andrade Barbosa, M.; Milori, D.M.B.P.; La Scala, N., Jr. Crop rotation and succession in a no-tillage system: Implications for CO₂ emission and soil attributes. *J. Environ. Manag.* **2019**, *245*, 8–15. [CrossRef]
58. Torres, C.M.M.E.; Jacovine, L.A.G.; Oliveira Neto, S.N.; Brianezi, D.; Alves, E.B. Agroforestry Systems in Brazil: An approach to carbon storage. *J. For. Res.* **2014**, *34*, 235–244. [CrossRef]
59. Mascarenhas, A.R.P.; Scoti, M.S.V.; Melo, R.R.; Corrêa, F.L.O.; Souza, E.F.M.; Andrade, R.A.; Beramin, A.C.; Muller, M.W. Atributos físicos e estoques de carbono do solo sob diferentes usos da terra em Rondônia, Amazônia Sul-Occidental. *J. For. Res.* **2017**, *37*, 19–27. [CrossRef]
60. Rocha, S.J.S.S.; Schettini, B.L.S.; Alves, E.B.N.; Villanova, P.H.; Torres, C.M.M.E.; Jacovine, L.A.G.; Oliveira Neto, S.N.; Brianezi, D. Balanço de carbono em três sistemas silvipastoril no sudeste do Brasil. *Rev. Espacios* **2017**, *38*, 1–8.
61. Torralba, M.; Fagerholm, N.; Burgess, P.J.; Moreno, G.; Plieninger, T. Do European agroforestry systems enhance biodiversity and ecosystem services? A meta-analysis. *Agric. Ecosyst. Environ.* **2016**, *230*, 150–161. [CrossRef]
62. Barbosa, M.A.; de Sousa Ferraz, R.L.; Coutinho, E.L.M.; Neto, A.M.C.; da Silva, M.S.; Fernandes, C.; Rigobelo, E.C. Multivariate analysis and modeling of soil quality indicators in long-term management systems. *Sci. Total Environ.* **2018**, *657*, 457–465. [CrossRef] [PubMed]
63. Lal, R. Soil carbon sequestration to mitigate climate change. *Geoderma* **2004**, *123*, 1–22. [CrossRef]
64. Ghorbani, M.; Amirahmadi, E.; Konvalina, P.; Moudrý, J.; Kopecký, M.; Hoang, T.N. Carbon pool dynamic and soil microbial respiration affected by land use alteration: A case study in humid subtropical area. *Land* **2023**, *12*, 459. [CrossRef]
65. Graf, A.; Herbst, M.; Weiermüller, L.; Huisman, J.A.; Prolingheuer, N.; Bornemann, L.; Vereecken, H. Analyzing the spatiotemporal variability of heterotrophic soil respiration at a field scale using orthogonal functions. *Geoderma* **2012**, *181*, 91–101. [CrossRef]

66. Silva, D.A.P.; Campos, M.C.C.; Mantovanelli, B.C.; Santos, A.C.S.; Soares, M.D.R.; Cunha, J.M. Variabilidade espacial da emissão de CO₂, temperatura e umidade do solo em área de pastagem na região Amazônica, Brasil. *Rev. Ciênc. Agrovet.* **2019**, *18*, 119–126. [CrossRef]
67. Maier, M.; Schack-Kirchner, H.; Hildebrand, E.E.; Schindler, D. Soil CO₂ efflux vs. soil respiration: Implications for flux models. *Agric. For. Meteorol.* **2011**, *151*, 1723–1730. [CrossRef]
68. Vargas-Terminel, M.L.; Flores-Rentería, D.; Sánchez-Mejía, Z.M.; Rojas-Robles, N.E.; Sandoval-Aguilar, M.; Chávez-Vergara, B.; Robles-Morua, A.; Garatuza-Payan, J.; Yépez, E.A. Soil Respiration Is Influenced by Seasonality, Forest Succession and Contrasting Biophysical Controls in a Tropical Dry Forest in Northwestern Mexico. *Soil Syst.* **2022**, *6*, 75. [CrossRef]
69. Saiz, G.; Green, C.; Butterbach-Bahl, K.; Kiese, R.; Avitabile, V.; Farrell, E.P. Seasonal and spatial variability of soil respiration in four Sitka spruce stands. *Plant Soil* **2006**, *287*, 161–176. [CrossRef]
70. Zhu, D.; Ciais, P.; Krinner, G.; Maignan, F.; Jornet Puig, A.; Hugelius, G. Controls of soil organic matter on soil thermal dynamics in the northern high latitudes. *Nat. Commun.* **2019**, *10*, 1–9. [CrossRef]
71. Conceição, M.; Manzatto, C.V.; Araújo, W.S.; Martin Neto, L.; Saab, S.C.; Cunha, T.J.F.; Freixo, A.A. *Estudo Comparativo de Métodos de Determinação do Teor de Matéria Orgânica em Organossolos*; Embrapa: Rio de Janeiro, Brazil, 1999; p. 4.
72. Painel Intergovernamental Sobre Mudanças Climáticas. An IPCC Special Report on Climate Change, Desertification, Land Degradation, Sustainable Land Management, Food Security, and Greenhouse Gas Fluxes in Terrestrial Ecosystems. 2019. Available online: <https://www.ipcc.ch/srccl/> (accessed on 3 March 2021).
73. Lal, R. Carbon sequestration. *Philos. Trans. R. Soc. Lond. Ser. B Biol. Sci.* **2008**, *363*, 815–830. [CrossRef] [PubMed]
74. Martins, M.R.; Cora, J.E.; Jorge, R.F.; Marcelo, A.V. Crop type influences soil aggregation and organic matter under no-tillage. *Soil Till. Res.* **2009**, *104*, 22–29. [CrossRef]
75. Silva-Olaya, A.M.; Cerri, C.E.P.; La Scala, N., Jr.; Dias, C.T.S.; Cerri, C.C. Carbon dioxide emissions under different soil tillage systems in mechanically harvested sugarcane. *Environ. Res. Lett.* **2013**, *8*, 1–8. [CrossRef]
76. Silva, L.J.; Oliveira, D.M.S.; Santos, R.S.; Oliveira, P.A.; Freitas, D.A.F.; Cherubin, M.R.; Cerri, C.E.P. Soil carbon dynamics in integrated agricultural systems in Minas Gerais state, Brazil: A meta-analysis. *Geoderma Reg.* **2024**, *36*, e00761. [CrossRef]
77. Longo, R.M.; Espíndola, C.R.; Ribeiro, A.L. Modificações na estabilidade de agregados no solo decorrentes da introdução de pastagens em áreas de Cerrado e Floresta Amazônica. *Rev. Bras. Eng. Agric. Ambient.* **2000**, *3*, 276–280. [CrossRef]
78. Hungria, M.; Franchini, J.C.; Brandão, O., Jr.; Kaschuk, G.; Souza, R.A. Soil microbial activity and crop sustainability in a long-term experiment with three tillage systems and two crop rotation systems. *Appl. Soil Ecol.* **2009**, *42*, 288–296. [CrossRef]
79. Shepherd, M.; Nichlos, S.; Selbie, D. An assessment of the role of soil organic matter in pasture resilience. *Grassl. Res. Pract. Ser.* **2021**, *17*, 179–190. [CrossRef]

Disclaimer/Publisher's Note: The statements, opinions and data contained in all publications are solely those of the individual author(s) and contributor(s) and not of MDPI and/or the editor(s). MDPI and/or the editor(s) disclaim responsibility for any injury to people or property resulting from any ideas, methods, instructions or products referred to in the content.